Project Report

BAN110:
Data preparation and Handling
-
census of income in US

Group Members

[redacted]

[redacted]

[redacted]

Vinchell Justin Dcosta

1. Dataset and task description

The dataset contains both numerical and categorical variables:

- Numerical Variables: There are 7 numerical variables in the dataset. These include: age, capital_gain, capital_loss, education_num, fnlwgt, hours_per_week.

- Categorical Variables: There are 8 categorical variables in the dataset. These include: education, income, marital_status, native_country, occupation, race, relationship, sex, workclass.

This dataset describes various demographic and socio-economic attributes related to individuals, potentially for use in social science or economics research, or in governmental statistics. It includes a mix of both numerical and categorical variables that together provide a comprehensive view of different aspects of the individuals represented in the data.

2. Loading data

```
libname project "/home/u63416215/BAN110/Project";

OPTIONS VALIDVARNAME = V7;
PROC IMPORT DATAFILE="/home/u63416215/BAN110/Project/adult.csv"
    DBMS= CSV
    OUT= project.census replace;
    GETNAMES=YES;
RUN;

/* Checking Data Contents */
Title "Contents of Stat Data";
PROC CONTENTS DATA= project.census;
RUN;
```

Contents of Stat Data

The CONTENTS Procedure

Data Set Name	PROJECT.CENSUS	**Observations**	32561
Member Type	DATA	**Variables**	15
Engine	V9	**Indexes**	0
Created	08/09/2023 12:53:40	**Observation Length**	152
Last Modified	08/09/2023 12:53:40	**Deleted Observations**	0
Protection		**Compressed**	NO
Data Set Type		**Sorted**	NO
Label			
Data Representation	SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_IA64		
Encoding	utf-8 Unicode (UTF-8)		

Engine/Host Dependent Information

Data Set Page Size	131072
Number of Data Set Pages	38
First Data Page	1
Max Obs per Page	861
Obs in First Data Page	836
Number of Data Set Repairs	0
Filename	/home/u63416215/BAN110/Project/census.sas7bdat
Release Created	9.0401M7
Host Created	Linux
Inode Number	16246166240
Access Permission	rw-r--r--
Owner Name	u63416215
File Size	5MB
File Size (bytes)	5111808

Alphabetic List of Variables and Attributes

#	Variable	Type	Len	Format	Informat
1	age	Num	8	BEST12.	BEST32.
11	capital_gain	Num	8	BEST12.	BEST32.
12	capital_loss	Num	8	BEST12.	BEST32.
4	education	Char	12	$12.	$12.
5	education_num	Num	8	BEST12.	BEST32.
3	fnlwgt	Num	8	BEST12.	BEST32.
13	hours_per_week	Num	8	BEST12.	BEST32.
15	income	Char	5	$5.	$5.
6	marital_status	Char	13	$13.	$13.
14	native_country	Char	13	$13.	$13.
7	occupation	Char	17	$17.	$17.
9	race	Char	5	$5.	$5.
8	relationship	Char	14	$14.	$14.
10	sex	Char	6	$6.	$6.
2	workclass	Char	16	$16.	$16.

Categorical variables

- Check and correct errors where necessary

```
/* Categorical Data */

title "Checking Categorical Variables in Cencus Dataset";
proc freq data= project.census;
table _character_ ;
RUN;
```

Checking Categorical Variables in Cencus Dataset

The FREQ Procedure

workclass	Frequency	Percent	Cumulative Frequency	Cumulative Percent
?	1836	5.64	1836	5.64
Federal-gov	960	2.95	2796	8.59
Local-gov	2093	6.43	4889	15.01
Never-worked	7	0.02	4896	15.04
Private	22696	69.70	27592	84.74
Self-emp-inc	1116	3.43	28708	88.17
Self-emp-not-inc	2541	7.80	31249	95.97
State-gov	1298	3.99	32547	99.96
Without-pay	14	0.04	32561	100.00

education	Frequency	Percent	Cumulative Frequency	Cumulative Percent
10th	933	2.87	933	2.87
11th	1175	3.61	2108	6.47
12th	433	1.33	2541	7.80
1st-4th	168	0.52	2709	8.32
5th-6th	333	1.02	3042	9.34
7th-8th	646	1.98	3688	11.33
9th	514	1.58	4202	12.91
Assoc-acdm	1067	3.28	5269	16.18
Assoc-voc	1382	4.24	6651	20.43
Bachelors	5355	16.45	12006	36.87
Doctorate	413	1.27	12419	38.14
HS-grad	10501	32.25	22920	70.39
Masters	1723	5.29	24643	75.68
Preschool	51	0.16	24694	75.84
Prof-school	576	1.77	25270	77.61
Some-college	7291	22.39	32561	100.00

marital_status	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Divorced	4443	13.65	4443	13.65
Married-AF-sp	23	0.07	4466	13.72
Married-civ-s	14976	45.99	19442	59.71
Married-spous	418	1.28	19860	60.99

marital_status	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Never-married	10683	32.81	30543	93.80
Separated	1025	3.15	31568	96.95
Widowed	993	3.05	32561	100.00

occupation	Frequency	Percent	Cumulative Frequency	Cumulative Percent
?	1843	5.66	1843	5.66
Adm-clerical	3770	11.58	5613	17.24
Armed-Forces	9	0.03	5622	17.27
Craft-repair	4099	12.59	9721	29.85
Exec-managerial	4066	12.49	13787	42.34
Farming-fishing	994	3.05	14781	45.39
Handlers-cleaners	1370	4.21	16151	49.60
Machine-op-inspct	2002	6.15	18153	55.75
Other-service	3295	10.12	21448	65.87
Priv-house-serv	149	0.46	21597	66.33
Prof-specialty	4140	12.71	25737	79.04
Protective-serv	649	1.99	26386	81.04
Sales	3650	11.21	30036	92.25
Tech-support	928	2.85	30964	95.10
Transport-moving	1597	4.90	32561	100.00

relationship	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Husband	13193	40.52	13193	40.52
Not-in-family	8305	25.51	21498	66.02
Other-relative	981	3.01	22479	69.04
Own-child	5068	15.56	27547	84.60
Unmarried	3446	10.58	30993	95.18
Wife	1568	4.82	32561	100.00

race	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Amer-	311	0.96	311	0.96
Asian	1039	3.19	1350	4.15
Black	3124	9.59	4474	13.74
Other	271	0.83	4745	14.57
White	27816	85.43	32561	100.00

sex	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Female	10771	33.08	10771	33.08
Male	21790	66.92	32561	100.00

native_country	Frequency	Percent	Cumulative Frequency	Cumulative Percent
?	583	1.79	583	1.79
Cambodia	19	0.06	602	1.85
Canada	121	0.37	723	2.22
China	75	0.23	798	2.45
Columbia	59	0.18	857	2.63
Cuba	95	0.29	952	2.92
Dominican-Rep	70	0.21	1022	3.14
Ecuador	28	0.09	1050	3.22
El-Salvador	106	0.33	1156	3.55
England	90	0.28	1246	3.83
France	29	0.09	1275	3.92
Germany	137	0.42	1412	4.34
Greece	29	0.09	1441	4.43
Guatemala	64	0.20	1505	4.62
Haiti	44	0.14	1549	4.76
Holand-Nether	1	0.00	1550	4.76
Honduras	13	0.04	1563	4.80
Hong	20	0.06	1583	4.86
Hungary	13	0.04	1596	4.90
India	100	0.31	1696	5.21
Iran	43	0.13	1739	5.34
Ireland	24	0.07	1763	5.41
Italy	73	0.22	1836	5.64
Jamaica	81	0.25	1917	5.89
Japan	62	0.19	1979	6.08
Laos	18	0.06	1997	6.13
Mexico	643	1.97	2640	8.11
Nicaragua	34	0.10	2674	8.21

native_country	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Outlying-US(G	14	0.04	2688	8.26
Peru	31	0.10	2719	8.35
Philippines	198	0.61	2917	8.96
Poland	60	0.18	2977	9.14
Portugal	37	0.11	3014	9.26
Puerto-Rico	114	0.35	3128	9.61
Scotland	12	0.04	3140	9.64
South	80	0.25	3220	9.89
Taiwan	51	0.16	3271	10.05
Thailand	18	0.06	3289	10.10
Trinadad&Toba	19	0.06	3308	10.16
United-States	29170	89.59	32478	99.75
Vietnam	67	0.21	32545	99.95
Yugoslavia	16	0.05	32561	100.00

income	Frequency	Percent	Cumulative Frequency	Cumulative Percent
<=50K	24720	75.92	24720	75.92
>50K	7841	24.08	32561	100.00

Categorical variables

- Check for missing values

```
/* Missing Character */
Data Project.Missing;
    SET Project.census;
    array Char_Vars {*} $_character_;
    do Loop = 1 to dim(Char_Vars);
        if Char_Vars {Loop} in ('?') then Char_Vars {Loop} = ' ';
    end;
    drop Loop;
Run;

title "Checking Missing Character Values";
proc format;
    value $Count_Missing ' '  = 'Missing'
                         other = 'Nonmissing';
run;

proc freq data= project.missing;
    tables _character_ / nocum missing;
    format _character_ $Count_Missing.;
run;
```

Checking Missing Character Values

The FREQ Procedure

workclass	Frequency	Percent
Missing	1836	5.64
Nonmissing	30725	94.36

education	Frequency	Percent
Nonmissing	32561	100.00

marital_status	Frequency	Percent
Nonmissing	32561	100.00

occupation	Frequency	Percent
Missing	1843	5.66
Nonmissing	30718	94.34

relationship	Frequency	Percent
Nonmissing	32561	100.00

race	Frequency	Percent
Nonmissing	32561	100.00

sex	Frequency	Percent
Nonmissing	32561	100.00

native_country	Frequency	Percent
Missing	583	1.79
Nonmissing	31978	98.21

income	Frequency	Percent
Nonmissing	32561	100.00

Checking Missing Character Values

Obs	age	workclass	fnlwgt	education	education_num	marital_status	occupation	relationship	race	sex	capital_gain	capital_loss	hours_per_week	native_country	income
167	30	Self-emp-not-inc	257295	Some-college	10	Never-married	Sales	Other-relative	Asian	Male	0	2258	40	South	<=50K
238	60		141221	Bachelors	13	Married-civ-s		Husband	Asian	Male	0	2163	25	South	<=50K
445	49	Private	137192	Bachelors	13	Married-civ-s	Sales	Husband	Asian	Male	0	1977	50	South	>50K
621	42	Federal-gov	108183	Masters	14	Married-civ-s	Prof-specialty	Husband	Asian	Male	0	1902	40	South	>50K
639	30	Self-emp-inc	173858	Some-college	10	Married-civ-s	Exec-managerial	Husband	Asian	Male	0	1902	40	South	>50K
995	20		114746	11th	7	Married-spous		Own-child	Asian	Female	0	1762	40	South	<=50K
1310	35	Private	140915	Bachelors	13	Never-married	Sales	Own-child	Asian	Male	0	1590	40	South	<=50K

Obs	age	workclass	fnlwgt	education	education_num	marital_status	occupation	relationship	race	sex	capital_gain	capital_loss	hours_per_week	native_country	income
1576	24		151153	Some-college	10	Never-married		Not-in-family	Asian	Male	99999	0	50	South	>50K
1897	54	Private	135803	Bachelors	13	Married-civ-s	Exec-managerial	Husband	Asian	Male	15024	0	60	South	>50K
1955	34	Self-emp-not-inc	198664	Bachelors	13	Married-civ-s	Craft-repair	Husband	Asian	Male	15024	0	70	South	>50K
1967	42	Private	119359	Prof-school	15	Married-civ-s	Sales	Wife	Amer-	Female	15024	0	40	South	>50K
3260	57	Private	182677	Bachelors	13	Married-civ-s	Exec-managerial	Husband	Asian	Male	4508	0	40	South	<=50K
4255	54		180211	Some-college	10	Married-civ-s		Husband	Asian	Male	0	0	60	South	>50K
5027	43	Private	158926	Masters	14	Married-civ-s	Prof-specialty	Wife	Asian	Female	0	0	50	South	<=50K
5928	45	Self-emp-not-inc	133969	HS-grad	9	Married-civ-s	Sales	Husband	Asian	Male	0	0	50	South	>50K
6141	34	Self-emp-not-inc	306982	Bachelors	13	Married-civ-s	Sales	Husband	Asian	Male	0	0	60	South	<=50K
6998	23	Private	128604	HS-grad	9	Never-married	Sales	Own-child	Asian	Male	0	0	48	South	<=50K
7753	36	Self-emp-not-inc	288585	HS-grad	9	Married-civ-s	Other-service	Wife	Asian	Female	0	0	20	South	<=50K
8015	33	Private	256211	HS-grad	9	Married-civ-s	Machine-op-inspct	Husband	Asian	Male	0	0	40	South	<=50K
8049	19		20469	HS-grad	9	Never-married		Other-relative	Asian	Female	0	0	12	South	<=50K
8846	58	Private	137547	Bachelors	13	Married-civ-s	Other-service	Husband	Asian	Male	0	0	40	South	<=50K
8930	32	Private	153078	Bachelors	13	Never-married	Other-service	Not-in-family	Asian	Male	0	0	40	South	<=50K
9577	31	Self-emp-inc	83748	Some-college	10	Married-civ-s	Exec-managerial	Wife	Asian	Female	0	0	70	South	<=50K
9875	50	Self-emp-not-inc	324506	HS-grad	9	Widowed	Exec-managerial	Unmarried	Asian	Female	0	0	48	South	<=50K
9939	28	Self-emp-not-inc	123983	Some-college	10	Married-civ-s	Sales	Own-child	Asian	Male	0	0	63	South	<=50K
10247	48	Private	118831	HS-grad	9	Divorced	Handlers-cleaners	Unmarried	Asian	Male	0	0	40	South	<=50K
10362	20	Self-emp-not-inc	112137	Some-college	10	Never-married	Prof-specialty	Other-relative	Asian	Female	0	0	20	South	<=50K
10364	45	Private	281565	HS-grad	9	Widowed	Other-service	Other-relative	Asian	Female	0	0	50	South	<=50K

Categorical variables

- Treat missing values

```
/* Treating errors */
/*South Korea only noted as South and Hong Kong as Hong */
proc print data= project.missing;
where find(native_country,'South',"i");
run;

Data project.country;
    Set project.missing;
if native_country = "South" then native_country = "South Korea";
if native_country = "Hong" then native_country = "Hong Kong";
Run;

title "Correcting Datapoints";
proc freq data= project.country;
    tables native_country / nocum missing;
run;

/* Treating missing value */
/* By deletion */
/* Choice of Deletion method: we could not assume missing value of country */
Data project.corrected_country;
    Set project.country;
if native_country = " " then delete;
Run;

title "Treating Missing Values with Deletion";
proc freq data= project.corrected_country;
    tables native_country / nocum;
run;
/* By imputation */
/* Choice of Imputation method: there is no unemployement recorded for occupation and workclass variable
so we assume that the missing value is considered as unemployed */
Data project.occupation;
    Set project.corrected_country;
if occupation= " " then occupation= "Unemployed";
if workclass= " " then workclass= "Unemployed";
Run;

title "Treating Missing Values with Imputation";
proc freq data= project.occupation;
    tables workclass occupation / nocum;
run;
```

Treating Missing Values with Deletion

The FREQ Procedure

native_country	Frequency	Percent
Cambodia	19	0.06
Canada	121	0.38
China	75	0.23
Columbia	59	0.18
Cuba	95	0.30
Dominican-Rep	70	0.22
Ecuador	28	0.09
El-Salvador	106	0.33
England	90	0.28
France	29	0.09
Germany	137	0.43
Greece	29	0.09
Guatemala	64	0.20
Haiti	44	0.14
Holand-Nether	1	0.00
Honduras	13	0.04
Hong Kong	20	0.06
Hungary	13	0.04
India	100	0.31
Iran	43	0.13
Ireland	24	0.08
Italy	73	0.23
Jamaica	81	0.25
Japan	62	0.19
Laos	18	0.06
Mexico	643	2.01
Nicaragua	34	0.11
Outlying-US(G	14	0.04
Peru	31	0.10
Philippines	198	0.62
Poland	60	0.19
Portugal	37	0.12
Puerto-Rico	114	0.36
Scotland	12	0.04
South Korea	80	0.25
Taiwan	51	0.16
Thailand	18	0.06
Trinadad&Toba	19	0.06
United-States	29170	91.22
Vietnam	67	0.21
Yugoslavia	16	0.05

Treating Missing Values with Imputation

The FREQ Procedure

workclass	Frequency	Percent
Federal-gov	943	2.95
Local-gov	2067	6.46
Never-worked	7	0.02
Private	22286	69.69
Self-emp-inc	1074	3.36
Self-emp-not-inc	2499	7.81
State-gov	1279	4.00
Unemployed	1809	5.66
Without-pay	14	0.04

occupation	Frequency	Percent
Adm-clerical	3721	11.64
Armed-Forces	9	0.03
Craft-repair	4030	12.60
Exec-managerial	3992	12.48
Farming-fishing	989	3.09
Handlers-cleaners	1350	4.22
Machine-op-inspct	1966	6.15
Other-service	3212	10.04
Priv-house-serv	143	0.45
Prof-specialty	4038	12.63
Protective-serv	644	2.01
Sales	3584	11.21
Tech-support	912	2.85
Transport-moving	1572	4.92

occupation	Frequency	Percent
Unemployed	1816	5.68

Categorical variables

- Create one or more derived variables

```
/* Derived Variable */
/* Creating new variable of the education level into bigger category, edu_group.
This category would give simpler analysis */
proc format;
    value $education "preschool", "1st-4th", "5th-6th", "7th-8th", "9th", "10th", "11th", "12th" = "Below High School"
                     "HS-grad" = "High School"
                     "Some-college", "Bachelors", "Masters", "Doctorate", "Prof-school" = "College/University"
                     other = 'Other';
run;

Data project.stat;
    Set project.occupation;
edu_group= education;
format edu_group $education.;
RUN;

title "Derived Education Variable to Edu_group";
proc freq data= project.stat;
    tables edu_group/ nocum;
run;
```

Derived Education Variable to Edu_group

The FREQ Procedure

edu_group	Frequency	Percent
Below High School	4119	12.88
Other	2471	7.73
College/University	15020	46.97
High School	10368	32.42

Numerical variables

- Check the Distribution of Some Numerical Variables

```
/*1 - Check the Distribution of Some Numerical Variables to Decide Which Method to Use for Outlier Detection:*/

proc univariate data=project.stat;
var age 'capital.gain'n 'capital.loss'n 'education.num'n fnlwgt 'hours.per.week'n  ;
run;
```

- Age: Mean: 38.58 Std Deviation: 13.64 Median: 37 Range: 73
- Capital Gain: Mean: 1077.65 Std Deviation: 7385 Median: 0 Range: 99999
- Capital Loss: Mean: 87.30 Std Deviation: 402.96 Median: 0 Range: 4356
- Education Number: Mean: 10.08 Std Deviation: 2.57 Median: 10 Range: 15
- Final Weight (fnlwgt): Mean: 189778.4 Std Deviation: 105550 Median: 178356 Range: 1472420
- Hours per Week: Mean: 40.44 Std Deviation: 12.35 Median: 40 Range: 98

Each section provides a statistical summary of the given variable, highlighting key metrics such as the mean, standard deviation, median, and range. Other statistics like skewness, kurtosis, and specific quantiles offer additional insights into the distribution and characteristics of the data.

Numerical variables

- Show the Statistics (Min, Max, Mean) and the Shape of the Distribution: */

```
/*Show the Statistics (Min, Max, Mean) and the Shape of the Distribution: */
proc means data=project.stat min max mean;
   var age 'capital.gain'n 'capital.loss'n 'education.num'n fnlwgt 'hours.per.week'n;
run;

proc sgplot data=project.stat;
   histogram 'capital.gain'n;
run;
```

The MEANS Procedure

Variable	Minimum	Maximum	Mean
age	17.0000000	90.0000000	38.5816468
capital.gain	0	99999.00	1077.65
capital.loss	0	4356.00	87.3038297
education.num	1.0000000	16.0000000	10.0806793
fnlwgt	12285.00	1484705.00	189778.37
hours.per.week	1.0000000	99.0000000	40.4374559

Numerical variables

- Correct errors by imputation or deletion.
  Detect and remove outliers.

```
/*2 - Correct Errors by Imputation or Deletion */

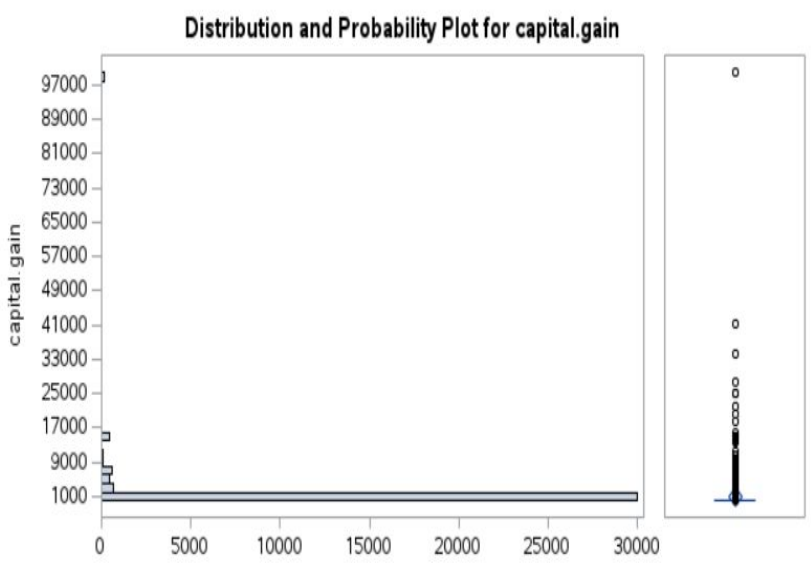
/* Assuming the imputation of median for negative values in 'capital.gain'n */
proc stdize data=project.stat out=project.corrected_data method=median reponly missing=0;
var capital_gain;
where capital_gain < 0;
run;
/* Choice of Imputation Method: The median is a robust statistic that is not affected by outliers.
If the distribution of capital_gain is skewed or contains extreme values,
imputing with the median might be more appropriate than using the mean. */
```

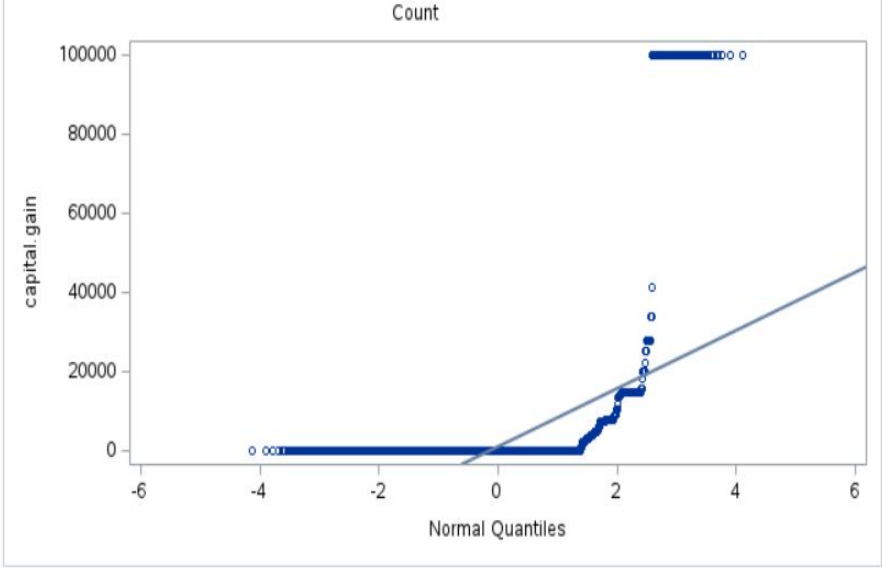
```
/*Detect and Remove Outliers*/
/* Assuming removal of outliers based on Z-score method for capital_gain */
data project.no_outliers;
   set project.corrected_data;
   if abs(capital_gain - mean) > 3 * std then delete;
run;
```

Numerical variables

- Test for normality and plot histogram and QQ plots for a variable with a skewed distribution.

```
/* Test for Normality and Plot Histogram and QQ Plots: */
proc univariate data=project.no_outliers plot;
   var 'capital.gain'n;
   histogram 'capital.gain'n / normal;
run;
```

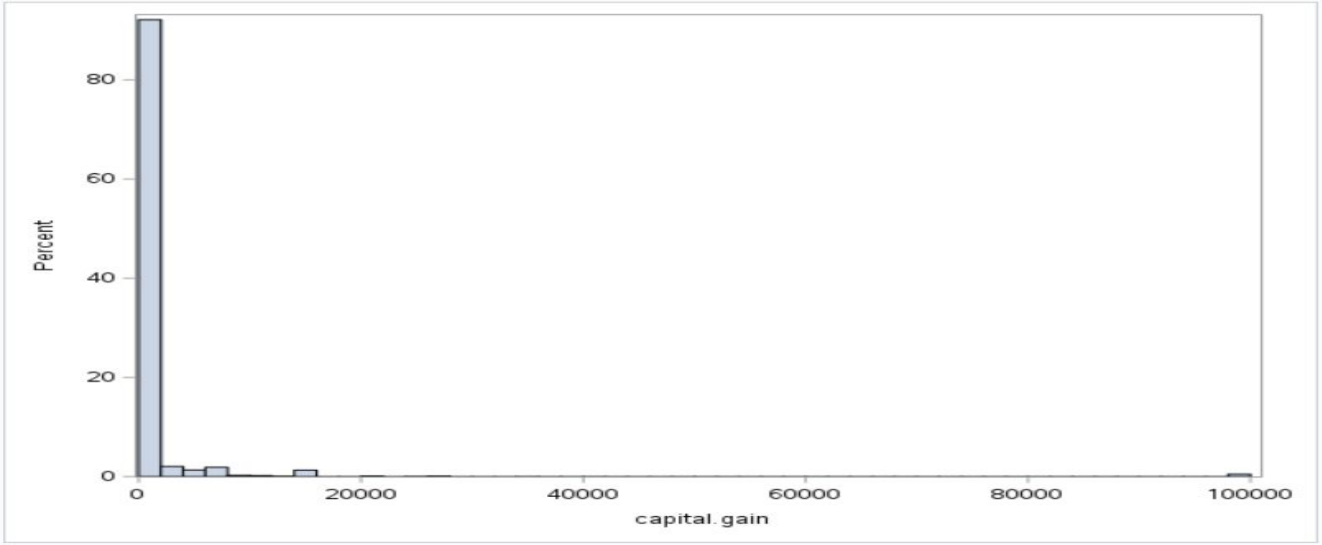




Numerical variables

- Apply a transformation and test for normality again with histogram and QQ plot.

```
39 proc means data=project.stat min max mean;
40    var age 'capital.gain'n 'capital.loss'n 'education.num'n fnlwgt 'hours.per.week'n;
41 run;
42
43 proc sgplot data=project.stat;
44    histogram 'capital.gain'n;
45 run;
```



Thank you